

Derivative of Matrices and Vectors 矩阵与向量的导数

1 标量函数的导数

1.1 引入：标量函数对标量的导数

回忆标量函数对标量的求导方法，对于标量一元函数 (scalar function) (1.1)

$$f(u) = u^2 - 2u + 1 \quad (1.1)$$

求导得：

$$\frac{df(u)}{du} = 2u - 2 \quad (1.2)$$

考虑标量二元函数的求导

$$f(u_1, u_2) = u_1^3 - u_2^2 + 2u_1 + 3u_2 + 1 \quad (1.3)$$

$$\begin{cases} \frac{\partial f(u_1, u_2)}{\partial u_1} = 3u_1^2 + 2 \\ \frac{\partial f(u_1, u_2)}{\partial u_2} = -2u_2 + 3 \end{cases} \quad (1.4)$$

1.2 标量函数对向量的导数

将 u_1, u_2 组合成向量 $\mathbf{u} = (u_1, u_2)^T$ ，那么 $f(u_1, u_2)$ 可以表示为 $f(\mathbf{u})$ 则式 (1.4) 可以写成标量函数对向量的导数，即

$$\frac{\partial f(\mathbf{u})}{\partial \mathbf{u}} = \begin{bmatrix} \frac{\partial f(\mathbf{u})}{\partial u_1} \\ \frac{\partial f(\mathbf{u})}{\partial u_2} \end{bmatrix} = \begin{bmatrix} 3u_1^2 + 2 \\ -2u_2 + 3 \end{bmatrix} \quad (1.5)$$

由此可见，标量函数对向量的导数，本质上就是标量函数对向量的各个分量求偏导，得到的就是函数 f 的梯度 ∇f 。

将上式推广到更高维度，考虑 n 维向量 $\mathbf{u} = (u_1, u_2, \dots, u_n)^T \in \mathbb{R}^n$,

$$\nabla f(\mathbf{u}) = \frac{\partial f(\mathbf{u})}{\partial \mathbf{u}} \triangleq \begin{bmatrix} \frac{\partial f(\mathbf{u})}{\partial u_1} \\ \frac{\partial f(\mathbf{u})}{\partial u_2} \\ \vdots \\ \frac{\partial f(\mathbf{u})}{\partial u_n} \end{bmatrix} \quad (1.6)$$

式 (1.6) 得到的 $n \times 1$ 矩阵共有 n 行，与分母部分 $\mathbf{u}$ 的行数相同，称为分母布局 (denominator layout)。

$$\frac{\partial f(\mathbf{u})}{\partial \mathbf{u}} \triangleq \begin{bmatrix} \frac{\partial f(\mathbf{u})}{\partial u_1} & \frac{\partial f(\mathbf{u})}{\partial u_2} & \dots & \frac{\partial f(\mathbf{u})}{\partial u_n} \end{bmatrix} \quad (1.7)$$

式 (1.7) 得到的 $1 \times n$ 矩阵共有 1 行，与分子部分 $f(\mathbf{u})$ 的行数相同，称为分子布局 (numerator layout)。

不难发现，分母布局与分子布局互为转置，二者没有本质区别，只是表示方式的不同。本笔记之后的推导将会统一采用分母布局。

1.3 标量函数对矩阵求导

假设变量 $\mathbf{X}$ 是一个 $m \times n$ 的矩阵， $f(\mathbf{X})$ 是关于变量 $\mathbf{X}$ 的标量函数，使用分母布局，定义标量函数对矩阵的导数为

$$\frac{\partial f(\mathbf{X})}{\partial \mathbf{X}} \triangleq \begin{bmatrix} \frac{\partial f(\mathbf{X})}{\partial x_{11}} & \frac{\partial f(\mathbf{X})}{\partial x_{12}} & \cdots & \frac{\partial f(\mathbf{X})}{\partial x_{1n}} \\ \frac{\partial f(\mathbf{X})}{\partial x_{21}} & \frac{\partial f(\mathbf{X})}{\partial x_{22}} & \cdots & \frac{\partial f(\mathbf{X})}{\partial x_{2n}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f(\mathbf{X})}{\partial x_{m1}} & \frac{\partial f(\mathbf{X})}{\partial x_{m2}} & \cdots & \frac{\partial f(\mathbf{X})}{\partial x_{mn}} \end{bmatrix} \quad (1.8)$$

1.4 总结

标量函数对向量的导数，本质上就是标量函数对向量的各个分量求偏导，再按照分母布局或者分子布局排列成矩阵。

同理，标量函数对矩阵的导数，本质上就是标量函数对矩阵的各个分量求偏导，再按照分母布局或者分子布局排列成矩阵。

2 向量函数的导数

2.1 向量函数对标量的导数

假设向量函数 $\mathbf{f}(u) = [f_1(u) \ f_2(u) \ \cdots \ f_m(u)]^T \in \mathbb{R}^m$ ，其中 u 为标量 $u \in \mathbb{R}$ ，对 u 求导数，得

$$\frac{\partial \mathbf{f}(u)}{\partial u} = \begin{bmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_2}{\partial u} & \cdots & \frac{\partial f_m}{\partial u} \end{bmatrix} \quad (2.1)$$

(这里为了与后面的符号一致，求导时采用了偏导的符号，但实际上这里不是求偏导)

式 (2.1) 得到的是 $1 \times m$ 矩阵，采用了分母布局。

2.2 向量函数对向量的导数

假设 $\mathbf{m}$ 维向量函数 $\mathbf{f}(\mathbf{u}) = [f_1(\mathbf{u}) \ f_2(\mathbf{u}) \ \cdots \ f_m(\mathbf{u})]^T \in \mathbb{R}^m$ ，其中 $\mathbf{u}$ 是 $\mathbf{n}$ 维向量 $\mathbf{u} \in \mathbb{R}^n$ ，对 $\mathbf{u}$ 求导数，有 2 种顺序：

2.2.1 顺序 1：先按照向量函数对标量求导，再按照标量函数对向量求导

Step1: 先按照向量函数对标量求导

先把分母部分 $\mathbf{u}$ 当作标量进行求导

$$\frac{\partial \mathbf{f}(\mathbf{u})}{\partial \mathbf{u}} = \begin{bmatrix} \frac{\partial f_1(\mathbf{u})}{\partial \mathbf{u}} & \frac{\partial f_2(\mathbf{u})}{\partial \mathbf{u}} & \cdots & \frac{\partial f_m(\mathbf{u})}{\partial \mathbf{u}} \end{bmatrix} \quad (2.2)$$

Step2: 再按照标量函数对向量求导

式 (2.2) 中，每一个分量 $\frac{\partial f_i(\mathbf{u})}{\partial \mathbf{u}}$, ($i = 1, 2, \dots, m$) 的分子部分都是标量函数，接下来把 $\mathbf{u}$ 作为向量，就可以按照标量函数对向量求导。

$$\frac{\partial f_i(\mathbf{u})}{\partial \mathbf{u}} = \begin{bmatrix} \frac{\partial f_i(\mathbf{u})}{\partial u_1} \\ \frac{\partial f_i(\mathbf{u})}{\partial u_2} \\ \vdots \\ \frac{\partial f_i(\mathbf{u})}{\partial u_n} \end{bmatrix} \quad (2.3)$$

综上，

$$\frac{\partial \mathbf{f}(\mathbf{u})}{\partial \mathbf{u}} = \begin{bmatrix} \frac{\partial f_1(\mathbf{u})}{\partial \mathbf{u}} & \frac{\partial f_2(\mathbf{u})}{\partial \mathbf{u}} & \cdots & \frac{\partial f_m(\mathbf{u})}{\partial \mathbf{u}} \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1(\mathbf{u})}{\partial u_1} & \frac{\partial f_2(\mathbf{u})}{\partial u_1} & \cdots & \frac{\partial f_m(\mathbf{u})}{\partial u_1} \\ \frac{\partial f_1(\mathbf{u})}{\partial u_2} & \frac{\partial f_2(\mathbf{u})}{\partial u_2} & \cdots & \frac{\partial f_m(\mathbf{u})}{\partial u_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_1(\mathbf{u})}{\partial u_n} & \frac{\partial f_2(\mathbf{u})}{\partial u_n} & \cdots & \frac{\partial f_m(\mathbf{u})}{\partial u_n} \end{bmatrix} \quad (2.4)$$

2.2.2 顺序 2：先按照标量函数对向量求导，再按照向量函数对标量求导

Step1: 先按照标量函数对向量求导

先把分子部分 $\mathbf{f}(\mathbf{u})$ 当作标量函数进行求导

$$\frac{\partial \mathbf{f}(\mathbf{u})}{\partial \mathbf{u}} = \begin{bmatrix} \frac{\partial \mathbf{f}(\mathbf{u})}{\partial u_1} \\ \frac{\partial \mathbf{f}(\mathbf{u})}{\partial u_2} \\ \vdots \\ \frac{\partial \mathbf{f}(\mathbf{u})}{\partial u_n} \end{bmatrix} \quad (2.5)$$

Step2: 再按照向量函数对标量求导

式 (2.5) 中，每一个分量 $\frac{\partial \mathbf{f}(\mathbf{u})}{\partial u_j}$, ($j = 1, 2, \dots, n$) 的分母部分都是标量，接下来只要把 $\mathbf{f}(\mathbf{u})$ 作为向量函数，就可以按照向量函数对标量求导

$$\frac{\partial \mathbf{f}(\mathbf{u})}{\partial u_j} = \begin{bmatrix} \frac{\partial f_1(\mathbf{u})}{\partial u_j} & \frac{\partial f_2(\mathbf{u})}{\partial u_j} & \cdots & \frac{\partial f_m(\mathbf{u})}{\partial u_j} \end{bmatrix} \quad (2.6)$$

综上,

$$\frac{\partial \mathbf{f}(\mathbf{u})}{\partial \mathbf{u}} = \begin{bmatrix} \frac{\partial \mathbf{f}(\mathbf{u})}{\partial u_1} \\ \frac{\partial \mathbf{f}(\mathbf{u})}{\partial u_2} \\ \vdots \\ \frac{\partial \mathbf{f}(\mathbf{u})}{\partial u_n} \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1(\mathbf{u})}{\partial u_1} & \frac{\partial f_2(\mathbf{u})}{\partial u_1} & \cdots & \frac{\partial f_m(\mathbf{u})}{\partial u_1} \\ \frac{\partial f_1(\mathbf{u})}{\partial u_2} & \frac{\partial f_2(\mathbf{u})}{\partial u_2} & \cdots & \frac{\partial f_m(\mathbf{u})}{\partial u_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_1(\mathbf{u})}{\partial u_n} & \frac{\partial f_2(\mathbf{u})}{\partial u_n} & \cdots & \frac{\partial f_m(\mathbf{u})}{\partial u_n} \end{bmatrix} \quad (2.7)$$

2.3 总结

两种顺序求得的结果一致, 都是

$$\frac{\partial \mathbf{f}(\mathbf{u})}{\partial \mathbf{u}} \triangleq \begin{bmatrix} \frac{\partial f_1(\mathbf{u})}{\partial u_1} & \frac{\partial f_2(\mathbf{u})}{\partial u_1} & \cdots & \frac{\partial f_m(\mathbf{u})}{\partial u_1} \\ \frac{\partial f_1(\mathbf{u})}{\partial u_2} & \frac{\partial f_2(\mathbf{u})}{\partial u_2} & \cdots & \frac{\partial f_m(\mathbf{u})}{\partial u_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_1(\mathbf{u})}{\partial u_n} & \frac{\partial f_2(\mathbf{u})}{\partial u_n} & \cdots & \frac{\partial f_m(\mathbf{u})}{\partial u_n} \end{bmatrix} \quad (2.8)$$

该结果是分母布局下的结果。

它也可以写成分子布局

$$\frac{\partial \mathbf{f}(\mathbf{u})}{\partial \mathbf{u}} \triangleq \begin{bmatrix} \frac{\partial f_1(\mathbf{u})}{\partial u_1} & \frac{\partial f_1(\mathbf{u})}{\partial u_2} & \cdots & \frac{\partial f_1(\mathbf{u})}{\partial u_n} \\ \frac{\partial f_2(\mathbf{u})}{\partial u_1} & \frac{\partial f_2(\mathbf{u})}{\partial u_2} & \cdots & \frac{\partial f_2(\mathbf{u})}{\partial u_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m(\mathbf{u})}{\partial u_1} & \frac{\partial f_m(\mathbf{u})}{\partial u_2} & \cdots & \frac{\partial f_m(\mathbf{u})}{\partial u_n} \end{bmatrix} \quad (2.9)$$

式 (2.9) 被称为雅可比矩阵 (**Jacobi matrix**)。

本笔记之后的推导将会统一采用分母布局。

3 导数和梯度

前面是根据导数的类型分别讨论, 下面从更一般的角度来看以上的导数。

3.1 定义

定义 3.1: 可微的定义

假设 $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, $x \in \text{dom} f$, 如果存在矩阵 $Df(x) \in \mathbb{R}^{m \times n}$ 满足

$$\lim_{\|h\| \rightarrow 0} \frac{\|f(x+h) - f(x) - Df(x)h\|}{\|h\|} = 0$$

则称函数 f 在 x 处可微, 矩阵 $Df(x)$ 称为 f 在 x 处的导数 (或 **Jacobian 矩阵**)。

由此可以得到导数的定义,

定义 3.2: 导数的定义

假设 $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, $x \in \text{dom} f$, 函数 f 可微, 那么函数 f 在 x 处的导数 (或 **Jacobian** 矩阵) $Df(x)$ 定义为

$$Df(x) = \lim_{\|h\| \rightarrow 0} \frac{\|f(x+h) - f(x)\|}{\|h\|}$$

其中, 每一个分量为

$$(Df(x))_{ij} = \frac{\partial f_i(x)}{\partial x_j}, \quad i = 1, 2, \dots, m, \quad j = 1, 2, \dots, n$$

所以向量函数对向量的导数的分子布局, 就是这里定义的一般的导数。

函数 f 在 x 处的一次逼近 (类似于“切线或切平面”) 为

$$f(y) = f(x) + Df(x)(y - x)$$

定义 3.3: 梯度的定义

假设 $f: \mathbb{R}^n \rightarrow \mathbb{R}$, $x \in \text{dom} f$, 如果 f 在 x 处可微, 则称 f 在 x 处的导数 $Df(x)$ 的转置为 f 在 x 处的**梯度**, 记作 $\nabla f(x)$ 。

$$\nabla f(x) = (Df(x))^T$$

这是一个列向量, $\nabla f(x) \in \mathbb{R}^n$ 。其中, 每一个分量为

$$(\nabla f(x))_j = \frac{\partial f(x)}{\partial x_j}, \quad j = 1, 2, \dots, n$$

所以标量函数对向量的导数的分母布局, 就是这里定义的一般的梯度。

函数 f 在 x 处的一次逼近 (类似于“切线或切平面”) 为

$$f(y) = f(x) + \nabla f(x)^T(y - x)$$

3.2 根据定义求导

下面从用几个例子来说明如何根据定义求导数。

例 1 考虑二次函数 $f: \mathbb{R}^n \rightarrow \mathbb{R}$,

$$f(x) = \frac{1}{2}x^T A x + b^T x + c$$

其中, $A \in \mathbb{R}^{n \times n}$, $b \in \mathbb{R}^n$, $c \in \mathbb{R}$ 为常数。求 f 的梯度 $\nabla f(x)$ 。

解 1

$$\begin{aligned}
 f(x+h) - f(x) &= \frac{1}{2}(x+h)^T A(x+h) + b^T(x+h) + c - \left(\frac{1}{2}x^T A x + b^T x + c \right) \\
 &= \frac{1}{2}(x^T A h + h^T A x + h^T A h) + b^T h \\
 &= \left(\frac{1}{2}(A + A^T)x + b \right)^T h + \frac{1}{2}h^T A h
 \end{aligned} \tag{3.1}$$

根据导数的定义, 有

$$Df(x) = \lim_{\|h\| \rightarrow 0} \frac{\|f(x+h) - f(x)\|}{\|h\|}$$

结合式 (3.1), 可得

$$Df(x) = \lim_{\|h\| \rightarrow 0} \frac{\| \left(\frac{1}{2}(A + A^T)x + b \right)^T h + \frac{1}{2}h^T A h \|}{\|h\|}$$

因为 $\frac{\|\frac{1}{2}h^T A h\|}{\|h\|}$ 是关于 h 的高阶无穷小, 所以

$$Df(x) = \left(\frac{1}{2}(A + A^T)x + b \right)^T$$

故

$$\nabla f(x) = (Df(x))^T = \frac{1}{2}(A + A^T)x + b$$

特别地, 当 A 为对称矩阵时, $\nabla f(x) = Ax + b$ 。

4 链式法则

4.1 引入: 标量函数对标量求导的链式法则

首先, 回忆标量函数对标量求导的链式法则。

根据图 1, 可以利用链式法则写出

$$\begin{cases} \frac{\partial J}{\partial u_1} = \frac{\partial J}{\partial y_1} \frac{\partial y_1}{\partial u_1} + \frac{\partial J}{\partial y_2} \frac{\partial y_2}{\partial u_1} + \frac{\partial J}{\partial y_3} \frac{\partial y_3}{\partial u_1} \\ \frac{\partial J}{\partial u_2} = \frac{\partial J}{\partial y_1} \frac{\partial y_1}{\partial u_2} + \frac{\partial J}{\partial y_2} \frac{\partial y_2}{\partial u_2} + \frac{\partial J}{\partial y_3} \frac{\partial y_3}{\partial u_2} \end{cases} \tag{4.1}$$

4.2 标量函数对向量求导的链式法则

将 (4.1) 写成矩阵形式

$$\begin{bmatrix} \frac{\partial J}{\partial u_1} \\ \frac{\partial J}{\partial u_2} \end{bmatrix} = \begin{bmatrix} \frac{\partial y_1}{\partial u_1} & \frac{\partial y_2}{\partial u_1} & \frac{\partial y_3}{\partial u_1} \\ \frac{\partial y_1}{\partial u_2} & \frac{\partial y_2}{\partial u_2} & \frac{\partial y_3}{\partial u_2} \end{bmatrix} \begin{bmatrix} \frac{\partial J}{\partial y_1} \\ \frac{\partial J}{\partial y_2} \\ \frac{\partial J}{\partial y_3} \end{bmatrix} \tag{4.2}$$

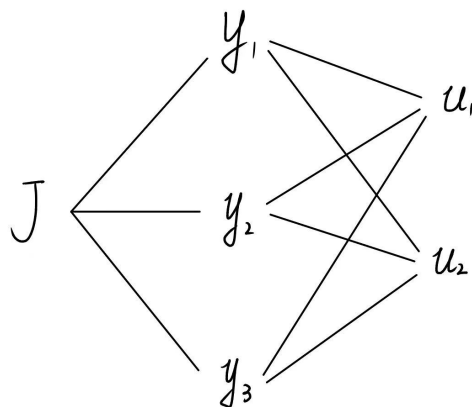


图 1 链式法则

假设 $\mathbf{u} = [u_1 \ u_2]^T$, $\mathbf{y}(\mathbf{u}) = [y_1(\mathbf{u}) \ y_2(\mathbf{u}) \ y_3(\mathbf{u})]^T$, 则依据分母布局, 有

$$\frac{\partial J}{\partial \mathbf{u}} = \begin{bmatrix} \frac{\partial J}{\partial u_1} \\ \frac{\partial J}{\partial u_2} \end{bmatrix}, \quad \frac{\partial \mathbf{y}}{\partial \mathbf{u}} = \begin{bmatrix} \frac{\partial y_1}{\partial u_1} & \frac{\partial y_1}{\partial u_2} & \frac{\partial y_1}{\partial u_3} \\ \frac{\partial y_2}{\partial u_1} & \frac{\partial y_2}{\partial u_2} & \frac{\partial y_2}{\partial u_3} \\ \frac{\partial y_3}{\partial u_1} & \frac{\partial y_3}{\partial u_2} & \frac{\partial y_3}{\partial u_3} \end{bmatrix}, \quad \frac{\partial J}{\partial \mathbf{y}} = \begin{bmatrix} \frac{\partial J}{\partial y_1} \\ \frac{\partial J}{\partial y_2} \\ \frac{\partial J}{\partial y_3} \end{bmatrix}$$

从而式 (4.2) 可以进一步写成

$$\frac{\partial J}{\partial \mathbf{u}} = \frac{\partial \mathbf{y}}{\partial \mathbf{u}} \frac{\partial J}{\partial \mathbf{y}} \quad (4.3)$$

将 $\mathbf{u}$ 推广到 $n \times 1$ 向量, $\mathbf{y}$ 推广到 $m \times 1$ 向量, 此式仍然成立。式 (4.3) 就是标量函数对向量求导的链式法则。

定理 4.1: 标量函数对向量求导的链式法则

若 $\mathbf{u} = [u_1 \ u_2 \ \dots \ u_n]^T \in \mathbb{R}^n$, $\mathbf{y}(\mathbf{u}) = [y_1(\mathbf{u}) \ y_2(\mathbf{u}) \ \dots \ y_m(\mathbf{u})]^T \in \mathbb{R}^m$, 有标量函数 $J = f(\mathbf{y}(\mathbf{u}))$, 则

$$\frac{\partial J}{\partial \mathbf{u}} = \frac{\partial \mathbf{y}}{\partial \mathbf{u}} \frac{\partial J}{\partial \mathbf{y}}$$

5 常用的矩阵求导公式

矩阵求导公式 5.1

If $\mathbf{u} = [u_1 \ u_2 \ \dots \ u_n]^T \in \mathbb{R}^n$ and $\mathbf{f} = [f_1 \ f_2 \ \dots \ f_n]^T \in \mathbb{R}^n$, then

$$\frac{\partial(\mathbf{u}^T \mathbf{f})}{\partial \mathbf{u}} = \frac{\partial(\mathbf{f}^T \mathbf{u})}{\partial \mathbf{u}} = \mathbf{f}$$

证明 1

$$\mathbf{u}^T \mathbf{f} = \begin{bmatrix} u_1 & u_2 & \dots & u_n \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{bmatrix} = u_1 f_1 + u_2 f_2 + \dots + u_n f_n \quad (5.1)$$

根据式 (1.6), 可得

$$\frac{\partial(\mathbf{u}^T \mathbf{f})}{\partial \mathbf{u}} = \begin{bmatrix} \frac{\partial(u_1 f_1 + u_2 f_2 + \dots + u_n f_n)}{\partial u_1} \\ \frac{\partial(u_1 f_1 + u_2 f_2 + \dots + u_n f_n)}{\partial u_2} \\ \vdots \\ \frac{\partial(u_1 f_1 + u_2 f_2 + \dots + u_n f_n)}{\partial u_n} \end{bmatrix} = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{bmatrix} = \mathbf{f} \quad (5.2)$$

因为

$$\mathbf{u}^T \mathbf{f} = \mathbf{f}^T \mathbf{u}$$

所以

$$\frac{\partial(\mathbf{f}^T \mathbf{u})}{\partial \mathbf{u}} = \frac{\partial(\mathbf{u}^T \mathbf{f})}{\partial \mathbf{u}} = \mathbf{f} \quad (5.3)$$

矩阵求导公式 5.2

If $\mathbf{u} = \begin{bmatrix} u_1 & u_2 & \dots & u_n \end{bmatrix}^T \in \mathbb{R}^n$ and $\mathbf{A}$ is a n by n matrix, then

$$\frac{\partial(\mathbf{A}\mathbf{u})}{\partial \mathbf{u}} = \mathbf{A}^T$$

证明 2 记矩阵 $\mathbf{A}$ 行向量的转置分别为

$$\mathbf{a}_1 = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \end{bmatrix}^T,$$

$$\mathbf{a}_2 = \begin{bmatrix} a_{21} & a_{22} & \dots & a_{2n} \end{bmatrix}^T,$$

...

$$\mathbf{a}_n = \begin{bmatrix} a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}^T$$

那么矩阵 $\mathbf{A}$ 可以表示为

$$\mathbf{A} = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_n^T \end{bmatrix}$$

从而可以把 $\mathbf{A}\mathbf{u}$ 写为

$$\mathbf{A}\mathbf{u} = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_n^T \end{bmatrix} \mathbf{u} = \begin{bmatrix} \mathbf{a}_1^T \mathbf{u} \\ \mathbf{a}_2^T \mathbf{u} \\ \vdots \\ \mathbf{a}_n^T \mathbf{u} \end{bmatrix} \quad (5.4)$$

将 $\mathbf{u}$ 当作标量, 利用式 (2.1), 得:

$$\frac{\partial \mathbf{A}\mathbf{u}}{\partial \mathbf{u}} = \begin{bmatrix} \frac{\partial \mathbf{a}_1^T \mathbf{u}}{\partial \mathbf{u}} & \frac{\partial \mathbf{a}_2^T \mathbf{u}}{\partial \mathbf{u}} & \cdots & \frac{\partial \mathbf{a}_n^T \mathbf{u}}{\partial \mathbf{u}} \end{bmatrix} \quad (5.5)$$

再利用矩阵求导公式 5.1, 对于 $i = 1, 2, \dots, n$, 有

$$\frac{\partial \mathbf{a}_i^T \mathbf{u}}{\partial \mathbf{u}} = \mathbf{a}_i \quad (5.6)$$

结合 (5.5) 和 (5.6), 得:

$$\frac{\partial \mathbf{A}\mathbf{u}}{\partial \mathbf{u}} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{bmatrix} = \mathbf{A}^T \quad (5.7)$$

矩阵求导公式 5.3

If $\mathbf{u} = \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix}^T \in \mathbb{R}^n$ and $\mathbf{A}$ is a n by n matrix, then

$$\frac{\partial (\mathbf{u}^T \mathbf{A}\mathbf{u})}{\partial \mathbf{u}} = \mathbf{A}\mathbf{u} + \mathbf{A}^T \mathbf{u}$$

Specially, if $\mathbf{A}$ is a symmetric matrix, which means $\mathbf{A} = \mathbf{A}^T$, so

$$\frac{\partial (\mathbf{u}^T \mathbf{A}\mathbf{u})}{\partial \mathbf{u}} = 2\mathbf{A}\mathbf{u}$$

证明 3

$$\mathbf{u}^T \mathbf{A}\mathbf{u} = \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$$

利用矩阵乘法公式可得

$$\begin{aligned}
\mathbf{u}^T \mathbf{A} \mathbf{u} &= \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \\
&= u_1 a_{11} u_1 + u_2 a_{21} u_1 + \cdots + u_n a_{n1} u_1 \\
&\quad + u_1 a_{12} u_2 + u_2 a_{22} u_2 + \cdots + u_n a_{n2} u_2 \\
&\quad + \cdots \\
&\quad + u_1 a_{1n} u_n + u_2 a_{2n} u_n + \cdots + u_n a_{nn} u_n
\end{aligned} \tag{5.8}$$

显然 $\mathbf{u}^T \mathbf{A} \mathbf{u}$ 是一个标量，因此 $\frac{\partial(\mathbf{u}^T \mathbf{A} \mathbf{u})}{\partial \mathbf{u}}$ 是标量函数对向量求导，根据式 (1.6) 和 (5.8)

$$\begin{aligned}
\frac{\partial(\mathbf{u}^T \mathbf{A} \mathbf{u})}{\partial \mathbf{u}} &= \begin{bmatrix} \frac{\partial(\mathbf{u}^T \mathbf{A} \mathbf{u})}{\partial u_1} \\ \frac{\partial(\mathbf{u}^T \mathbf{A} \mathbf{u})}{\partial u_2} \\ \vdots \\ \frac{\partial(\mathbf{u}^T \mathbf{A} \mathbf{u})}{\partial u_n} \end{bmatrix} \\
&= \begin{bmatrix} u_1 a_{11} + u_2 a_{21} + \cdots + u_n a_{n1} + a_{11} u_1 + a_{12} u_2 + \cdots + a_{1n} u_n \\ u_1 a_{12} + u_2 a_{22} + \cdots + u_n a_{n2} + a_{21} u_1 + a_{22} u_2 + \cdots + a_{2n} u_n \\ \vdots \\ u_1 a_{1n} + u_2 a_{2n} + \cdots + u_n a_{nn} + a_{n1} u_1 + a_{n2} u_2 + \cdots + a_{nn} u_n \end{bmatrix} \\
&= \begin{bmatrix} a_{11} u_1 + a_{12} u_2 + \cdots + a_{1n} u_n \\ a_{21} u_1 + a_{22} u_2 + \cdots + a_{2n} u_n \\ \vdots \\ a_{n1} u_1 + a_{n2} u_2 + \cdots + a_{nn} u_n \end{bmatrix} + \begin{bmatrix} u_1 a_{11} + u_2 a_{21} + \cdots + u_n a_{n1} \\ u_1 a_{12} + u_2 a_{22} + \cdots + u_n a_{n2} \\ \vdots \\ u_1 a_{1n} + u_2 a_{2n} + \cdots + u_n a_{nn} \end{bmatrix} \\
&= \mathbf{A} \mathbf{u} + \mathbf{A}^T \mathbf{u}
\end{aligned} \tag{5.9}$$

矩阵求导公式 5.4

If $\mathbf{u} = \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix}^T \in \mathbb{R}^n$ and $\mathbf{A}$ is a n by n matrix, then

$$\frac{\partial^2(\mathbf{u}^T \mathbf{A} \mathbf{u})}{\partial \mathbf{u}^2} = \mathbf{A} + \mathbf{A}^T$$

Specially, if $\mathbf{A}$ is a symmetric matrix, which means $\mathbf{A} = \mathbf{A}^T$, so

$$\frac{\partial^2(\mathbf{u}^T \mathbf{A} \mathbf{u})}{\partial \mathbf{u}^2} = 2\mathbf{A}$$

证明 4 由矩阵求导公式 5.3 可知

$$\begin{aligned}\frac{\partial^2(\mathbf{u}^T \mathbf{A} \mathbf{u})}{\partial \mathbf{u}^2} &= \frac{\partial}{\partial \mathbf{u}} \frac{\partial \mathbf{u}^T \mathbf{A} \mathbf{u}}{\partial \mathbf{u}} \\ &= \frac{\partial}{\partial \mathbf{u}} (\mathbf{A} \mathbf{u} + \mathbf{A}^T \mathbf{u}) \\ &= \frac{\partial \mathbf{A} \mathbf{u}}{\partial \mathbf{u}} + \frac{\partial \mathbf{A}^T \mathbf{u}}{\partial \mathbf{u}}\end{aligned}\quad (5.10)$$

结合式矩阵求导公式 5.2, 得

$$\frac{\partial \mathbf{A} \mathbf{u}}{\partial \mathbf{u}} = \mathbf{A}^T, \quad \frac{\partial \mathbf{A}^T \mathbf{u}}{\partial \mathbf{u}} = \mathbf{A} \quad (5.11)$$

将 (5.11) 代入 (5.10), 得到

$$\frac{\partial^2(\mathbf{u}^T \mathbf{A} \mathbf{u})}{\partial \mathbf{u}^2} = \mathbf{A}^T + \mathbf{A} \quad (5.12)$$

矩阵求导公式 5.5

If $\mathbf{u} = \begin{bmatrix} u_1 & u_2 & \dots & u_n \end{bmatrix}^T \in \mathbb{R}^n$, $\mathbf{A}$ is a m by n matrix, $\mathbf{y}(\mathbf{u}) = \mathbf{A} \mathbf{u}$ and $\mathbf{B}$ is a m by m symmetric matrix, $J = \mathbf{y}^T \mathbf{B} \mathbf{y}$, then

$$\frac{\partial J}{\partial \mathbf{u}} = 2\mathbf{A}^T \mathbf{B} \mathbf{y}$$

证明 5 根据定理 4.1

$$\frac{\partial J}{\partial \mathbf{u}} = \frac{\partial \mathbf{y}}{\partial \mathbf{u}} \frac{\partial J}{\partial \mathbf{y}} = \frac{\partial \mathbf{y}}{\partial \mathbf{u}} \frac{\partial (\mathbf{y}^T \mathbf{B} \mathbf{y})}{\partial \mathbf{y}} \quad (5.13)$$

由矩阵求导公式 5.2 可知

$$\frac{\partial \mathbf{y}}{\partial \mathbf{u}} = \mathbf{A}^T \quad (5.14)$$

由矩阵求导公式 5.3 可知

$$\frac{\partial (\mathbf{y}^T \mathbf{B} \mathbf{y})}{\partial \mathbf{y}} = 2\mathbf{B} \mathbf{y} \quad (5.15)$$

将 (5.14) 和 (5.15) 代入 (5.13), 得

$$\frac{\partial J}{\partial \mathbf{u}} = 2\mathbf{A}^T \mathbf{B} \mathbf{y} \quad (5.16)$$

矩阵求导公式 5.6

If $\mathbf{A}$ is a n by n matrix and $\mathbf{X}$ is a n by n symmetric matrix, then

$$\frac{\partial \text{tr}(\mathbf{X} \mathbf{A}^T)}{\partial \mathbf{X}} = \mathbf{A}$$

证明 6

$$tr(\mathbf{XA}^T) = a_{11}x_{11} + a_{12}x_{12} + \cdots + a_{ij}x_{ij} + \cdots + a_{n(n-1)}x_{n(n-1)} + a_{nn}x_{nn} = \sum_{i=1}^n \sum_{j=1}^n a_{ij}x_{ij} \quad (5.17)$$

代入 (1.8), 可得

$$\frac{\partial tr(\mathbf{XA}^T)}{\partial \mathbf{X}} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} = \mathbf{A} \quad (5.18)$$

矩阵求导公式 5.7

If $\mathbf{A}$ is a n by n matrix and $\mathbf{X}$ is a n by n symmetric matrix, then

$$\frac{\partial tr(\mathbf{XAX}^T)}{\partial \mathbf{X}} = \mathbf{K}(\mathbf{A} + \mathbf{A}^T)$$

Specially, if $\mathbf{A}$ is a symmetric matrix, which means $\mathbf{A} = \mathbf{A}^T$, then

$$\frac{\partial tr(\mathbf{XAX}^T)}{\partial \mathbf{X}} = 2\mathbf{XA}$$

证明 7 略, 暴力展开即可得证。

参考文献

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